POKHARA UNIVERSITY

Level: Bachelor | Semester : Spring | Year : 2023
Programme: BE | Full Marks: 100
Course: Computer Architecture | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1.	a)	Differentiate between Computer Organization and Computer Architecture. Explain the functional view of the computer with reference to each component.	8
	b)	Define RTL. Explain about Bus and Register transfers in RTL.	7
2.	a)	Explain different types of registers used in Computer System with examples. What is the use of registers in Computer System?	7
	b)	Perform -15 *6 using Booth's algorithm.	8
3.	a)	Explain the restoring division algorithm with flowchart.	8
	b)	Explain the working principle of Micro-programmed implementation of control unit. Why is it slower than Hardwired Control Unit?	7
4.	a)	What is associative memory? How read /write operation takes place in associative memory?	7
	b)	As we know cache memory size is smaller than main memory, how is it possible to map main memory data into cache memory? Explain in detail.	8
5.	a)	How Interrupt driven I/O differ from programmed I/O? Explain with necessary diagram.	7
	b)	How instruction pipelining supports in parallelism? Explain.	8
6.	a)	Cache coherence problem arise in multiprocessor system and must be resolved. Justify this statement.	8
	b)	What are the performance issues that arise in multicore computers?	7
7.		Write short notes on: (Any two)	2×5
	a)	Control unit.	
	b)	RISC VS CISC	
	c)	Flynn's Classification	

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